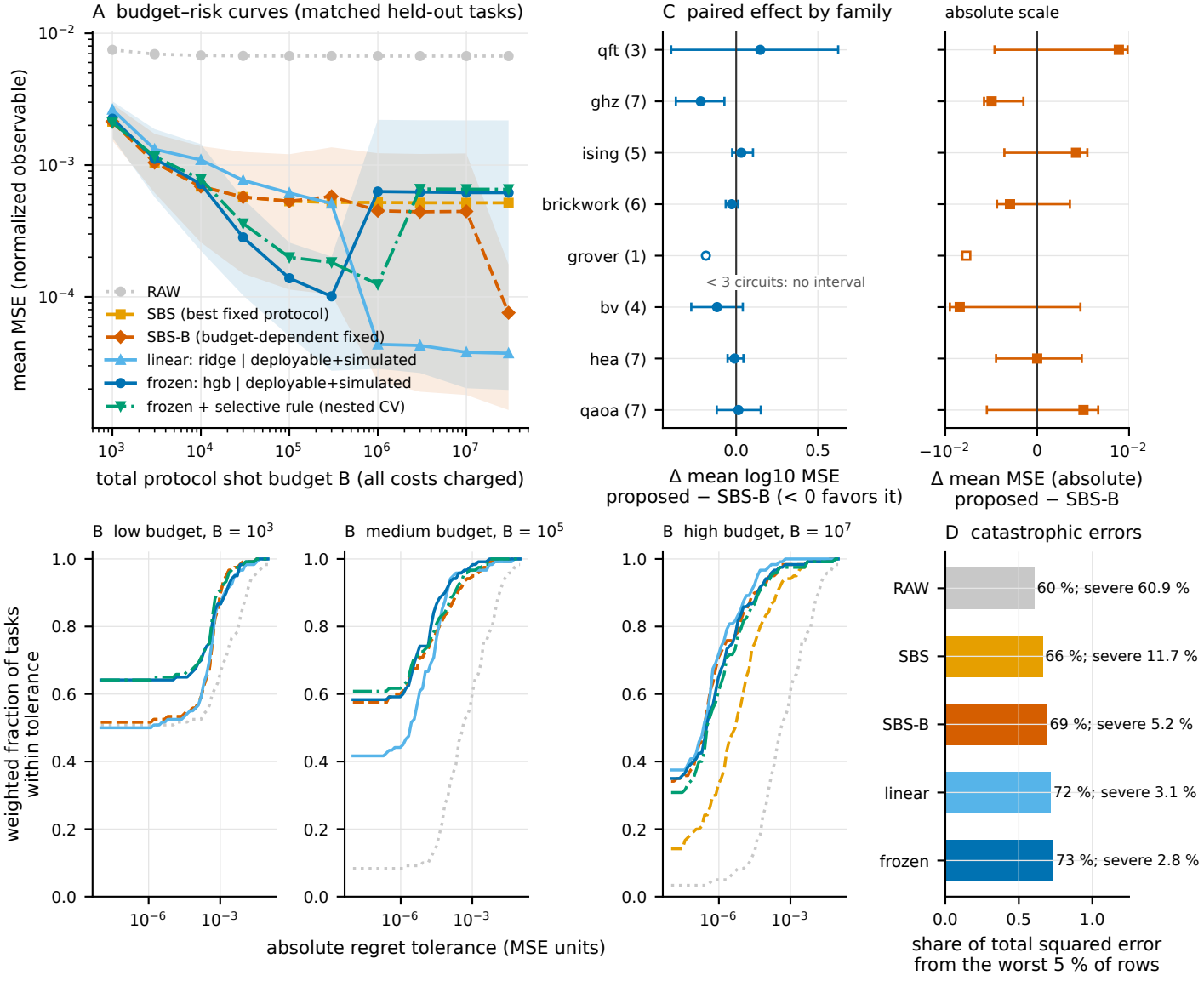
Simulated observations (emulated fake-device trials); no hardware data.
A budget–risk curves (matched held-out tasks)
mean MSE (normalized observable)
total protocol shot budget B (all costs charged)
RAW
SBS (best fixed protocol)
SBS-B (budget-dependent fixed)
linear: ridge | deployable+simulated
frozen: hgb | deployable+simulated
frozen + selective rule (nested CV)
C paired effect by family
qft (3)
ghz (7)
ising (5)
brickwork (6)
grover (1)
< 3 circuits: no interval
bv (4)
hea (7)
qaoa (7)
Δ mean log10 MSE
proposed − SBS-B (< 0 favors it)
absolute scale
Δ mean MSE (absolute)
proposed − SBS-B
B low budget, B = 10^3
B medium budget, B = 10^5
B high budget, B = 10^7
weighted fraction of tasks within tolerance
absolute regret tolerance (MSE units)
D catastrophic errors
RAW
SBS
SBS-B
linear
frozen
60 %; severe 60.9 %
66 %; severe 11.7 %
69 %; severe 5.2 %
72 %; severe 3.1 %
73 %; severe 2.8 %
share of total squared error
from the worst 5 % of rows